Recorded clean-clone verification

Repository source: CLEAN_CLONE_CHECK.md

Record type: Pre-release baseline retained with release v0.1.0
Release date: September 6, 2026
Canonical repository: https://github.com/swihart/minvol-degree3-spectral-bound

This record predates the final v0.1.0 release-candidate commit and is retained as reproducibility evidence. The exact v0.1.0 commit is tested again after all release metadata is committed and before the tag is created; that final result is recorded on the GitHub release page.

Result: PASS with the full Poppler PDF preflight delegated to GitHub Actions
UTC timestamp: 2026-09-04 10:05:19
Tested commit: `96cea57275070f21babf30477ab1b128ec0f5eb6`
Repository: `swihart/minvol-degree3-spectral-bound`

Purpose

This record documents a successful reproduction attempt from a newly cloned copy of the repository. The test was designed to detect hidden local files, uncommitted dependencies, machine-specific build assumptions, and generated outputs that differ from the versions tracked in Git.

The tested commit preceded this documentation commit. The result therefore applies exactly to commit `96cea57275070f21babf30477ab1b128ec0f5eb6`; later commits require their own verification when a final release candidate is prepared.

Recorded result

```
CLEAN CLONE CHECK: PASS WITH PDF PREFLIGHT DELEGATED TO GITHUB ACTIONS
Commit: 96cea57275070f21babf30477ab1b128ec0f5eb6
Repository: swihart/minvol-degree3-spectral-bound
PDF validation: basic local PDF and checksum checks passed; full PDF preflight delegated to GitHub
    Actions
Tracked working tree after all checks and rebuilds: clean
```

Machine-specific temporary-directory and artifact-directory paths have been omitted from this public-facing record. They were retained in the local ignored `ci-artifacts/` directory when the test was run.

Procedure completed

Starting from a clean checkout whose `HEAD` matched `origin/main`, the helper:

1. cloned the repository into a new temporary directory;
2. created an isolated Python virtual environment outside the clone;
3. installed the pinned Python dependencies;
4. ran every exact and numerical Python check;
5. ran both independent base-R checks;
6. rebuilt the LaTeX manuscript;
7. rebuilt every Markdown-derived PDF;
8. ran the metadata preflight;
9. performed local PDF-header and checksum validation; and
10. confirmed that all tracked files remained byte-for-byte unchanged after the checks and rebuilds.

The same tested commit also had all three GitHub Actions jobs green. The hosted document job supplied the full Poppler-based PDF preflight that was unavailable on the local Mac.

Environment

Component	Recorded value
Operating system	macOS 26.2, Darwin 25.2.0, `x86_64`
Git	2.50.1 (Apple Git-155)
Python	3.13.15
NumPy	2.3.5
SymPy	1.14.0
R	4.6.0
Pandoc	2.19.2
XeTeX	3.141592653-2.6-0.999994, TeX Live 2022
`latexmk`	4.77
Poppler tools	Not installed locally; full preflight delegated to GitHub Actions

The clean result is particularly useful because the local document toolchain differed from the pinned Ubuntu CI toolchain. Despite those differences, the tracked manuscript, rendered documentation PDFs, checksum manifest, and recorded verification outputs were reproduced without changing a tracked byte.

Interpretation and limits

This test supports the following claims:

- the committed computational checks run successfully from a fresh clone;
- the committed document sources rebuild successfully in the recorded local environment;
- the regenerated tracked outputs match the committed outputs byte for byte;
- no uncommitted local source file was needed for the successful run; and
- the full PDF preflight also passed in the clean hosted GitHub Actions environment.

It does **not** constitute independent subject-matter review, peer review, or a certification that the proposed mathematical theorem is correct. The current status remains:

> AI-assisted, non-peer-reviewed research draft seeking independent mathematical verification.

See `REPRODUCIBILITY.md` for the reproduction procedure and `VERIFICATION_STATUS.md` for the boundary between computational reproducibility and mathematical verification.